Digital Bank - Poori Feature Guide

Customer, Bank staff (Internal officer, Field officer) aur Admin ke saare features. Har feature ke saath: **kya karta hai**, **kaise chalana hai (steps)** aur **screen pe kya dikhega (output)**.

1. Project chalana aur login
2. Customer ke features
3. Internal officer (bank staff) ke features
4. Field officer (external employee) ke features
5. Admin - Banking pages
6. Admin - Operations pages (pods, simulation, rules)
7. Sir ke liye demos (saga, RabbitMQ, service down)
8. Aam masle aur hal

1. Project chalana aur login

Backend aur front chalana

1. **Docker Desktop** on karo. Project folder mein `docker compose up -d` chalao (SQL Server, RabbitMQ, Redis).
2. Visual Studio mein `DigitalBanking.slnx` kholo. Solution pe right click → **Configure Startup Projects** → **Multiple startup projects** → 9 projects (Gateway, Identity, Customer, Account, Transaction, Notification, Audit, Admin, Chat) ko **Start** karo.
3. **Ctrl + F5** dabao (bina debug). Is se ek service band karne pe baaki band nahi hoti.
4. Front: `src\Web\digital-banking-web` mein `npm start`, phir browser mein `http://localhost:4200`.

> **Output:** Login page khulega "Welcome back - Sign in to Digital Bank".

Demo accounts

Kaun	Email	Password	Kahan se login
Customer (Demo Customer)	customer@digitalbank.local	Customer@12345	Customer login
Customer (Ali Raza)	ali.raza@digitalbank.local	Customer@12345	Customer login
Customer (Sara Khan)	sara.khan@digitalbank.local	Customer@12345	Customer login
Internal officer (Ayesha)	internal@digitalbank.local	Internal@12345	Bank staff portal
Field officer (Omar)	external@digitalbank.local	External@12345	Bank staff portal
Admin	admin@digitalbank.local	Admin@12345	Bank staff portal

Login

1. Customer: login page pe email/password daalo → **Login**. Ya **Quick demo login** dabao aur account pe tap karo.
2. Staff/Admin: neeche **Bank staff portal** → email/password → **Enter portal**.

> **Output:** Customer → Home (balance card). Internal/Field officer → "Internal desk" / "Field desk". Admin → "Operations command center".

> Staff account customer page pe daloge to laal toast: "This app is for customers only". Customer account staff portal pe daloge to bhi error.

Test karne ka tareeqa

Do browser windows kholo: ek normal, ek Incognito (Ctrl+Shift+N). Ek mein customer, doosri mein staff/admin ya doosra customer. Ek taraf kaam karo, doosri taraf live asar dekho.

2. Customer ke features

Neeche 5 tab: **Home, Accounts, Transfer (beech wala), Support, More**. Har screen pe upar patli patti mein customer ka **naam aur email** dikhta hai. Har kaam ka result upar **toast** mein aata hai (hara = success, laal = error).

Naya customer banna (Open account)

Sirf profile banti hai, bank account nahi. Account ke liye alag request bhejni hoti hai.

1. Login page → **Open account**.
2. Full name, Email, CNIC, Mobile, Address, Password (kam az kam 8) bharo → **Create account**.

> **Output:** Seedha Home khul jata hai. Card pe "No account yet - Request an account". Notification aati hai "Welcome to Digital Banking". KYC = Pending.

Bank account ki request bhejna

Request staff ke paas jati hai. Approve hone pe hi account khulta hai.

1. **Accounts** tab → upar **+** (ya Request account).
2. **Savings** ya **Current** chuno, Purpose likho → **Next: documents**.
3. CNIC front aur back ki photo (JPG/PNG, 8MB se kam) → **Next: signature**.
4. Box mein signature karo → **Next: declaration** → checkbox tick → **Submit for approval**.

> **Output:** Hara toast "Request sent. The bank will activate your account...". Accounts mein "In review - Pending". Home pe "In review" card.

> Ek waqt mein sirf ek Pending request. Reject ho to "Request declined" aur **Send again** button aata hai.

Paise daalna / nikalna (Add money, Cash out)

1. **Accounts** tab → account card pe **Add money** → amount → **Deposit**.
2. Nikalne ke liye **Cash out** → amount → **Withdraw**.

> **Output:** Hara toast "Amount added." / "Amount withdrawn." aur balance foran update. Notification bhi aati hai.

> Balance kam ho to laal toast "Insufficient account balance." Account Frozen ho to "This account is frozen."

Account number copy karna

1. **Accounts** tab → card pe number ke saath gol **copy icon** dabao.

> **Output:** Poora account number clipboard pe copy. Icon 2 second ke liye ✓ ban jata hai. Ye number dost ko bhejo taake wo tumhe paise bhej sake.

Paise bhejna (Transfer)

1. Neeche beech wala **Transfer** button.
2. **From account** chuno, **Beneficiary account** mein saamne wale ka account number (paste kar sakte ho), Amount, Purpose.
3. **Confirm transfer**.

> **Output: Success page**: bada hara ✓, "Transfer successful", amount, aur receipt (To, From, Purpose, Reference TXN..., Date). Buttons: **Done** (Home) aur **Send another**. Upar toast "Transfer sent". Receiver ke paas bina refresh toast "Transfer received" aur balance badal jata hai.

> Balance kam: laal toast, kuch nahi katta. Saamne wala account frozen: paise kat ke khud wapas (refund) - status Compensated.

Transfer jab service down ho (maintenance)

Kisi service ke band hone pe bhi transfer zaaya nahi hota.

1. **Transaction service band:** Confirm transfer dabao → popup "Service under maintenance" (koi paisa nahi kata). Do button: **Send anyway** aur **Cancel**.
2. **Send anyway** dabao → transfer RabbitMQ queue mein save.
3. **Account service band:** transfer khud save ho jata hai, popup ki zaroorat nahi.

> **Output: Sunehra (gold) page** "Transfer scheduled" + note "service maintenance mein hai, transfer save hai, wapas aate hi khud chala jayega". Service chalu hone ke kuch second baad dono ko toast ("Transfer sent" / "Transfer received") aur balance bina refresh update.

> Service 10 minute tak wapas na aaye to transfer cancel: paisa nahi katta, ya kata tha to refund. Balance kam nikla to History mein Failed.

Live update (bina refresh)

Koi bhi notification aaye (paise mile, account khula, request reject) to app khud update hota hai.

1. Ek window mein Ali login, doosri mein customer se Ali ko paise bhejo.

> **Output:** Ali ki screen pe upar hara toast "Transfer received: ..." aur Home/Accounts/History ka data khud badal jata hai. Refresh ki zaroorat nahi.

History (statement)

1. Home → **History** (ya More → Transaction history).
2. Search box mein amount/ref likho. Chips: **All / In / Out / Transfer**.
3. **Filter** → account aur From/To date → **Apply filter**.

> **Output:** Din ke hisaab se list, har row pe type, time, status, amount. Upar "Money in" aur "Money out" ka total.

Notifications

1. Home pe ghanti (bell) icon. Kisi notification pe tap → poori detail. **Read all** se sab read.

> **Output:** Welcome, Account opened, Deposit, Withdrawal, Transfer sent/received, Password changed waghera ki list.

Profile, photo, password

1. **More** → **My profile**: naam, mobile, address badlo → **Save details**. Photo pe tap karke photo lagao.
2. **More** → **Change password**: purana + naya → **Update password**.
3. Upar naam wali patti pe tap karne se bhi profile khulti hai.

Output: Toast "Profile updated." / "Password updated." Password badalne pe baaki devices se logout.

Photo sirf usi browser mein save hoti hai.

Password bhool gaye (Forgot password)

1. Login page → **Forgot password** → email → **Send reset code**.
2. Code (demo mein screen pe hi dikhta hai) + naya password → **Update password**.

Output: Toast "Password updated. Sign in with your new password." aur login page.

Support chat

1. **Support** tab → **Bank Support**. Message likho, file bhejo, voice note, ya voice/video call.

Output: Staff ka jawab live aata hai, ticks (sent/delivered/read), "typing..." aur online dot dikhte hain.

Customer sirf bank staff se chat kar sakta hai, doosre customer se nahi.

3. Internal officer (bank staff)

Kaam: **account request Approve/Reject** karna aur **KYC verify** karna. Login: Bank staff portal se `internal@digitalbank.local`. Neeche tab: **Home, People, Cards, History, Chat**.

Home (desk)

> **Output:** Bank ke numbers: Customers, Accounts, Pending requests, KYC wait, Frozen, aaj ki transactions, Money in. "Waiting for review" list aur shortcuts (Requests, Customers, History, Chat, Alerts, KYC).

Account request Approve / Reject (Cards)

1. **Cards** tab → **Requests** chip.
2. **View CNIC & sign** se documents dekho.
3. **Approve** / **Reject**. Rejected request pe **Undo reject**.

> **Output:** Approve: toast "Account approved and opened." Customer ka naya account (number "10..." se, balance 0, Active) ban jata hai aur customer ko live "Account opened" toast. Reject: "Request rejected." aur customer ko note "Documents need correction." Undo: "Request moved back to review."

KYC verify

1. Home → **KYC** → customer ke saath **Verify**.

> **Output:** Toast "KYC verified." Customer list se hat jata hai, status Verified.

> KYC undo nahi hoti. KYC ke baghair bhi account approve ho sakta hai.

People (customers dhoondhna)

1. **People** tab → naam/phone/email/CNIC se search, **All / Pending / Verified** filter.
2. Customer pe tap → **Message customer**.

> **Output:** Customer ki details, uske accounts aur balance. Message customer se seedha chat khulti hai.

History, Alerts, Chat

1. **History**: poore bank ki transactions (sirf dekhne ke liye).
2. **Alerts**: apni notifications.
3. **Chat**: customers ke support chat ka jawab, **+** se direct chat ya **New group**.

4. Field officer (external employee)

Ye **sirf dekh sakta hai aur chat kar sakta hai**. Login: `external@digitalbank.local`. Tab: **Home, People, Chat**. Upar "Field desk".

Kya kar sakta hai

1. **Home**: bank ke numbers.
2. **People**: customer search, details, **Message customer**.
3. **Chat**: customers aur staff se direct chat, groups.

Kaam	Field officer	Internal officer	Admin
Customer list dekhna	Haan	Haan	Haan
Request Approve / Reject	Nahi	Haan	Haan
KYC Verify	Nahi	Haan	Haan
Account Freeze / Unfreeze	Nahi	Nahi	Haan
Audit trail, Operations	Nahi	Nahi	Haan

5. Admin - Banking pages (asli bank ka data)

Login: `admin@digitalbank.local`. Left menu mein do hisse: **Banking** (asli data) aur **Operations** (simulation aur system control).

Bank overview

> **Output:** Customers, KYC pending, Accounts (frozen kitne), Applications waiting, aaj ki transactions, bank mein kul paise. Links: **Review**, **Open queue**, **Open ledger**.

Customers

1. Search ya **KYC pending** filter → customer pe click → **Verify KYC**.

> **Output:** Side drawer mein customer ki details; verify ke baad KYC = Verified.

Accounts (requests aur freeze)

1. **Applications** tab → request pe click → documents → **Approve & open account** / **Reject** / **Return to queue**.
2. **Live accounts** tab → account pe click → **Freeze account** / **Unfreeze account**.

> **Output:** Approve se account khulta hai. Freeze ke baad us account mein na paise aa sakte na ja sakte; usko bheja gaya paisa sender ko khud wapas.

Transactions, Audit trail, Chat

1. **Transactions**: saari transactions, Money in/out, sab se badi; click pe detail.
2. **Audit trail**: kis ne kab kya kiya (login, register, account khulna, deposit, transfer, password change). Click pe poora JSON.
3. **Chat**: bari screen wala chat.

6. Admin - Operations pages

Upar wali bar (har page pe)

1. **Start simulation / Pause**: nakli transactions chalao/roko.
2. **Reset**: checkbox chuno (Traffic rules, Chaos, Pods, Database) → **Reset selected**. Customer data delete nahi hota.
3. Sun/moon icon: dark/light theme.

> **Output:** Status pill "Simulation live" + TPS. Reset ke baad toast "Reset done: ...".

> Reset mein "Pods" tick ho to saari services ka Self-healing wapas ON ho jata hai.

Command center

1. Map pe province click → **Activity multiplier** slider (0-5x), **Surge 3x · 90s**, **Suspend 15 min** / **Resume region**.
2. **Base load** slider (5-1000 TPS).
3. **Scenarios**: Eid rush, Salary day, Bot attack, CPU pressure, DB storm, Slow database, Core banking errors.
4. **What needs attention**: masla + fix button (Block client, Rate limit, Scale to 2, Promote replica...).

> **Output:** Live TPS, success rate, latency, province map, chart, live transaction feed, "Why transactions fail".

Infrastructure - ek service band / chalu karna (pods)

Asli services (jo VS se chal rahi hain) ko yahan se band ya chalu kar sakte ho.

1. Compute section mein service ki row (jaise **transaction**).
2. Pehle **Self-healing OFF** karo, warna service khud wapas aa jayegi.
3. Band: pod pe **Kill**, ya **−** dabate jao jab tak 0.
4. Chalu: **+**, pod pe **Restart**, ya Self-healing ON.
5. **Cordon / Uncordon**: pod ko traffic se hatao/wapas lao.
6. Database node: **Take offline** (Auto failover on ho to replica primary ban jata hai), **Force failover**, **Planned switchover**, **Replication** on/off.

> **Output:** Pod card pe state (Running / Crashed), "N of M serving". Service 0 pe ho to gateway us ke liye 503 deta hai, baaki services chalti rehti hain. Chalu karne ke 5-10 second baad Running.

> VS mein F5 (debug) se chala rahe ho to ek kill se poora VS session band ho sakta hai. Demo ke liye Ctrl+F5 use karo. Admin se restart hui service VS ke andar nahi chalti (breakpoint nahi lagega).

Regions & customers

1. **Bulk add customers**: province, city, ginti (10K-10M) → **Add**.
2. **Seed by population**: poore mulk mein abaadi ke hisaab se.
3. **Find a customer**: naam/CNIC/mobile → **Search**.

> **Output:** Load jobs ki progress (rows/s, ETA), search results table aur har region ka "fan-out" time.

Load & health

1. Top clients → **Block** / **Limit 5/s**. Hot endpoints → **Disable**.
2. Chaos controls: Bot flood, CPU burn, Database pressure, Background traffic (toggle + slider).

> **Output:** Har service ka CPU, req/s, p95, errors; gateway traffic chart (success, errors, blocked).

Database explorer, Logs, Traffic rules

1. **Database explorer**: tables dekho, filter/sort, **SQL console** (read-only, 1000 rows tak).
2. **Logs explorer**: har request; search, time (5m-24h), row click → **Show only this client**, **Follow this trace**, **Block this client**.
3. **Traffic rules**: type chuno (Block client, Rate-limit, Suspend region, Disable endpoint, Throttle service) → value, expiry, reason → **Activate rule**. Hatane ke liye **Remove** / **Remove all**.

> **Output:** Rule 2 second mein lag jata hai; blocked request ko 403, rate limit pe 429. Login aur admin console kabhi block nahi hote.

7. Sir ke liye demos

RabbitMQ kholna

1. Browser: `http://localhost:15672`, login `guest / guest`.
2. **Exchanges** → `digitalbanking.events`. **Queues** → har service ki queue (notification, audit, transaction-service...).

> **Output:** Koi deposit/transfer karo to graph mein message aata aur consume hota dikhta hai.

Demo A: Saga refund (sab se aasaan)

1. Admin → Accounts → Live accounts → Ali ka account **Freeze**.
2. Customer se Ali ko transfer.

> **Output:** Laal toast "Transfer failed and the source account was refunded." Paise kat ke wapas. Admin → Transactions mein status **Compensated**.

Demo B: Notification service band, message queue mein ruka

1. Admin → Infrastructure → notification: Self-healing OFF → 0.
2. Customer deposit ya transfer karo (kaam karega).
3. RabbitMQ → Queues → notification queue ka **Ready** barhta hai.
4. Notification wapas **+**.

> **Output:** Queue khali ho jati hai aur notifications/toasts aa jate hain. Point: ek service band ho to bank chalta rehta hai, message lost nahi hota.

Demo C: Transaction service band → Send anyway → baad mein transfer

1. Admin → Infrastructure → transaction: Self-healing OFF → 0.
2. Customer se transfer → popup "Service under maintenance" → **Send anyway**.
3. Sunehra page "Transfer scheduled".
4. RabbitMQ → Queues → `transaction-service` mein **Ready: 1** (message ruka hua).
5. Admin se transaction **+**.

> **Output:** Queue 0 ho jati hai, dono windows mein toast ("Transfer sent" / "Transfer received") aur balance bina refresh badal jata hai.

> Demo se pehle transaction service ko naye code ke saath ek baar chalne do, tabhi uski queue RabbitMQ mein bind hoti hai.

Demo D: Account service band → transfer khud baad mein

1. Admin → Infrastructure → account: Self-healing OFF → 0.
2. Customer se transfer → seedha sunehra "Transfer scheduled" page.
3. Account wapas **+**.

> **Output:** 5-10 second mein transfer khud complete, dono ko live toast, balance update. (Background job har 5 second retry karta hai.)

Sir ko ek line mein

"Microservices RabbitMQ events se baat karti hain. Ek service down ho to message queue mein ruk jata hai aur service wapas aate hi process hota hai. Transfer Saga pattern se hota hai: beech mein fail ho to retry, aur na ho sake to paisa khud refund. Notification SignalR se live user tak pahunchti hai."

8. Aam masle aur hal

Masla	Hal
"Cannot reach the bank"	Backend (Gateway port 5000) chal raha hai? Login page → Server settings → Bank server mein `http://localhost:5000`.
Kill kiya lekin service khud wapas aa gayi	Service ka **Self-healing OFF** karo, phir Kill / 0.
Ek service kill ki to poora project band	VS mein F5 ki jagah **Ctrl+F5**, ya `.\start-all.ps1` se chalao.
Transfer dropdown mein account nahi	Customer ka account approve nahi hua. Staff → Cards → Approve.
Notification har ek 3 baar	Notification service InApp, Email, SMS teeno save karti hai (known behaviour).
Receiver ki History mein transfer nahi	History sirf apne bheje hue dikhati hai; receiver ko notification aur balance mein dikhta hai.
Code badla lekin asar nahi	VS mein saari services restart karo (Gateway, Transaction, Chat, Notification).